Negotiated Autonomy

How Personal AI Learns Its Scope of Action in Spatial Work

Jianxuan Shao | PhD Research Proposal | Revision 3 | 30 October 2026

Proposed supervisor: Prof. Can Liu

School of Creative Media, City University of Hong Kong

Abstract

An AI rearranges a workspace and moves a monitor its user expected to remain untouched. Understanding the intended layout did not establish permission for that action. I propose that most spatial grants have a compact scope: visible objects or a region, a permitted operation and a duration. I will test this representation and build a desktop/XR collaborator that negotiates grants through speech and learns defaults for repeated actions. Comparisons with an editable rule panel will measure authorisation effort alongside unwanted intervention. The research will identify where compact spatial permissions work, where they fail and how agreements change through use.

1 Motivation and Proposition

“Improve this workspace, but keep the window workstation” leaves room for disagreement about moving its monitor. “Move these chairs while I am away” explicitly delegates work; “not that one” narrows it. My goal is personal AI that connects conversation, contextual perception and execution through agreements people understand and revise.

The proposition that permission has geometry becomes testable as H1: at least 80% of grants in room-editing tasks have a scope expressible as an object set or region, an operation and a duration without extra condition fields. An omitted duration remains unresolved until clarified. Conditional grants, dependencies and exceptions are potential counterexamples, not information to hide inside an operation label. Execution mode is recorded separately.

2 Related Work and Research Gap

Reference-Based Manipulation formalises spatial reference through Source, Anchor and Frame [1]. EchoLadder provides reversible scene suggestions and documents how users inspect subtle changes [2]. This proposal extends control over individual suggestions into standing grants with expiry and a revisable history across sessions. It tests whether a compact scope record preserves what people actually authorise.

Orality [3], Rambler [7] and IntentPrism [8] support developing and inspecting intentions. Authorisation has an additional pragmatic function: “those chairs are awkward” expresses a judgement, whereas “move those chairs until I return” grants action. Study 1 will distinguish requests, grants, exceptions and retractions as speech acts tied to visible objects.

Horvitz's mixed-initiative principles address decisions under uncertainty [4]; Parasuraman et al. distinguish automation types and levels [5]; Amershi et al. guide correction and adaptation [6]. Sensible Agent adapts proactive AR assistance to context [9]. This project tests representational adequacy and negotiation cost: whether spatial grants fit the proposed fields, and whether dialogue with learned defaults reduces authorisation effort compared with explicit rules.

3 Research Questions

RQ1: How are spatial permissions granted and revised in speech and action, and which grants defeat the proposed compact representation?

RQ2: How do a rule panel, conversational authorisation and conversational authorisation with learned defaults differ in delegation overhead, unwanted actions and task completion?

RQ3: How do standing grants and preferred execution modes change across sessions, and when do authorised action and user expectations diverge?

4 Prototype and Mechanisms

4.1 Representing a grant

Speech and editor state ground proposed edits in a resettable room with a fixed asset library. For “move chairs on this side of the desk”, Source is the desk, Anchor its edge and Frame the user's viewpoint [1]; chairs are the targets. A gesture selects the region, while a floating label displays operation and expiry. A spatial model previews the result. Timestamped records distinguish the scene when permission was expressed from the scene at execution.

4.2 Commitment and continued work

The prototype offers suggestions, ghost previews, sandbox edits and commitment to the shared scene. Previews are discarded, sandbox changes reset and committed edits retain undo. Background work receives a separate time-limited grant. The extension to EchoLadder [2] is a standing agreement governing later actions, with a history recording who changed its scope and when.

4.3 Learning reusable defaults

For each user and object-operation class, an ordinal estimator learns the preferred default: suggest, preview, sandbox or commit. Confirmed approvals and corrections provide labels; silence does not. The user approves a standing grant and its maximum commitment level once, after which the learned default governs repeated actions within that grant. The benefit to test is fewer authorisation episodes, not merely fewer violations. Enlarging scope or its ceiling requires a new grant. The task record stores this permission history and the evidence behind mode changes.

4.4 Interrupting and returning control

A spoken stop or palm gesture cancels queued edits in the indicated scope. Direct manipulation pauses conflicting edits for scope review. Completed changes remain visible and undoable, and the interface identifies unfinished work before a renewed instruction resumes it.

5 Studies and Evaluation

Study 1: Speech acts and representational coverage. Fourteen participants will redesign workspaces with a Wizard-of-Oz assistant. Free-form work precedes introduction of the schema and targeted counterexamples. Two coders will test participant-confirmed grants against the fixed three-field schema. H1 uses free-form grants, with ambiguous cases reported separately; targeted cases diagnose failures. The outputs are a speech-act taxonomy, a coverage estimate and failure cases requiring richer permissions.

Study 2: Negotiation versus settings. Thirty participants will complete counterbalanced tasks with (A) an editable rule panel, (B) conversational authorisation and (C) conversational authorisation plus learned defaults. All share permission expressiveness, initial grants, scene displays, tools, stop and undo. Task requests use the same speech interface; only authorisation entry differs. A calibration task seeds the same defaults; only C updates them automatically during measured trials. A versus B tests expression and revision; B versus C tests learning. Rules remain editable throughout. I will preregister delegation overhead and unwanted actions as primary outcomes, model participant and task effects, and report completion alongside effect sizes and uncertainty.

Measure	Operational definition
Delegation overhead	Authorisation utterances, gestures, panel edits and active time; initial setup and later revisions separated.
Autonomy drift	Executed edits outside the latest explicit grant, coded by detection delay; unnoticed at 120 active-inspection seconds marks delayed drift.
Regret and repair	Unwanted actions, reason-coded undo, and time or edits required to restore intended work.
Reliance and completion	Predicted scope adherence versus observed adherence; user-endorsed completion, latency and checking effort.

The detection clock starts when a change becomes visible during active inspection, including after a return from absence. A change not yet displayed for inspection is unassessed, not automatically missed. I will retain continuous detection times and separately code regret, later preference changes and unwanted actions within valid grants. Grant violations audit grounding or execution failures; they need not occur for learning to reduce overhead. Model claims, tool success and user-goal achievement remain distinct.

Study 3: Agreements across sessions. Twelve participants will use the prototype for six weeks. Weekly event reviews will reconstruct permission histories, distinguishing deliberate expansion and withdrawal from unrecognised divergence. This deployment answers RQ3.

Preliminary Work

I built Game Fellows, a bounded companion, and prepared a counterbalanced Solo/Fellow comparison with paired seeds, event logs and perceived-control questions. Human evaluation is planned. In an image-detection capstone, I reconstructed 900 decisions from frozen outputs and coefficients, exposing class-specific trade-offs hidden by overall improvement. This supports reconstructing actions against their contemporaneous permissions.

6 Contributions and Schedule

The thesis will contribute an account of authorisation speech acts, evidence for the coverage and limits of compact spatial grants, and tested mechanisms for reducing delegation effort while tracking unwanted actions.

Period	Work and output
Year 1	Speech-act study; coverage test; desktop/XR prototype.
Year 2	Standing grants and mode learning; technical evaluation.
Year 3	Rule-panel and dialogue comparisons; overhead and action audit.
Year 4	Six-week deployment; synthesis and thesis.

Studies will obtain ethics approval and informed consent, minimise recorded data, and support withdrawal and deletion.

7 Fit and First Collaboration

Can Liu's work connects spatial reference [1], reversible scene design [2] and spoken thought [3]. My architectural training and CAVE installation experience provide practical grounding for testing that connection. In the first six months, I propose building a spatial permission mock-up and conducting the Wizard-of-Oz speech-act study, establishing which grants a compact representation preserves before implementing learned defaults.

References

[1] Yangyang He, Zhuangze Hou, Yonglin Chen and Can Liu. 2026. Reference-Based Manipulation: A Framework and Pipeline for Multimodal Spatial Reasoning. UIST 2026 (accepted). doi:10.1145/3830398.3830657

[2] Zhuangze Hou, Jingze Tian, Nianlong Li, Farong Ren and Can Liu. 2025. EchoLadder: Progressive AI-Assisted Design of Immersive VR Scenes. UIST 2025. doi:10.1145/3746059.3747659

[3] Wengxi Li, Jingze Tian and Can Liu. 2026. Orality: A Semantic Canvas for Externalizing and Clarifying Thoughts with Speech. CHI 2026. doi:10.1145/3772318.3791713

[4] Eric Horvitz. 1999. Principles of Mixed-Initiative User Interfaces. CHI 1999, 159-166. doi:10.1145/302979.303030

[5] Raja Parasuraman, Thomas B. Sheridan and Christopher D. Wickens. 2000. A Model for Types and Levels of Human Interaction with Automation. IEEE Transactions on Systems, Man, and Cybernetics - Part A: Systems and Humans 30(3), 286-297. doi:10.1109/3468.844354

[6] Saleema Amershi et al. 2019. Guidelines for Human-AI Interaction. CHI 2019. doi:10.1145/3290605.3300233

[7] Susan Lin et al. 2024. Rambler: Supporting Writing With Speech via LLM-Assisted Gist Manipulation. CHI 2024, Article 1043. doi:10.1145/3613904.3642217

[8] Zehuan Wang, Jiaqi Xiao, Jingwei Sun and Can Liu. 2025. IntentPrism: Human-AI Intent Manifestation for Web Information Foraging. CHI EA 2025, Article 345. doi:10.1145/3706599.3719744

[9] Geonsun Lee et al. 2025. Sensible Agent: A Framework for Unobtrusive Interaction with Proactive AR Agents. UIST 2025, Article 119. doi:10.1145/3746059.3747748